

Soil Health & Crop Recommendation Report

Report Date: November 06, 2025 at 21:04:41 UTC

Farmer Information	
Name:	rameshidliwada
Phone:	1234567890
Location:	pune
Land Size:	0.10 acres

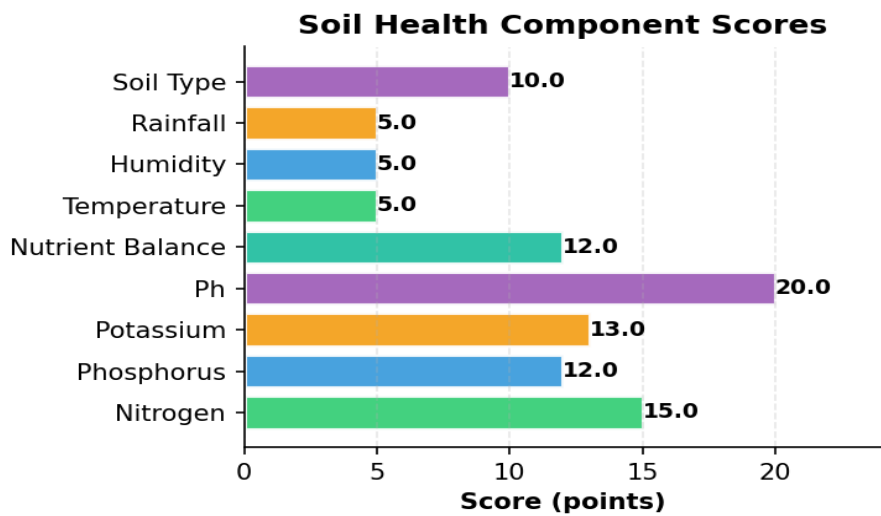
Parameter	Value	Unit
Nitrogen (N)	67.0	mg/kg
Phosphorus (P)	50.0	mg/kg
Potassium (K)	50.0	mg/kg
pH Level	6.50	
Temperature	25.0	°C
Humidity	60.0	%
Rainfall	123.0	mm
Soil Type	Alluvial Soil	

Soil Health Assessment

Overall Soil Health Score		97.0/100	Excellent
Component	Score	Points	
Nitrogen	15.0	points	
Phosphorus	12.0	points	
Potassium	13.0	points	
Ph	20.0	points	
Nutrient Balance	12.0	points	
Temperature	5.0	points	
Humidity	5.0	points	
Rainfall	5.0	points	
Soil Type	10.0	points	

Soil Health & Crop Recommendation System

Comprehensive Agricultural Analysis Report



Recommended Crop

Predicted Crop	Jute
Confidence Level	38.0%

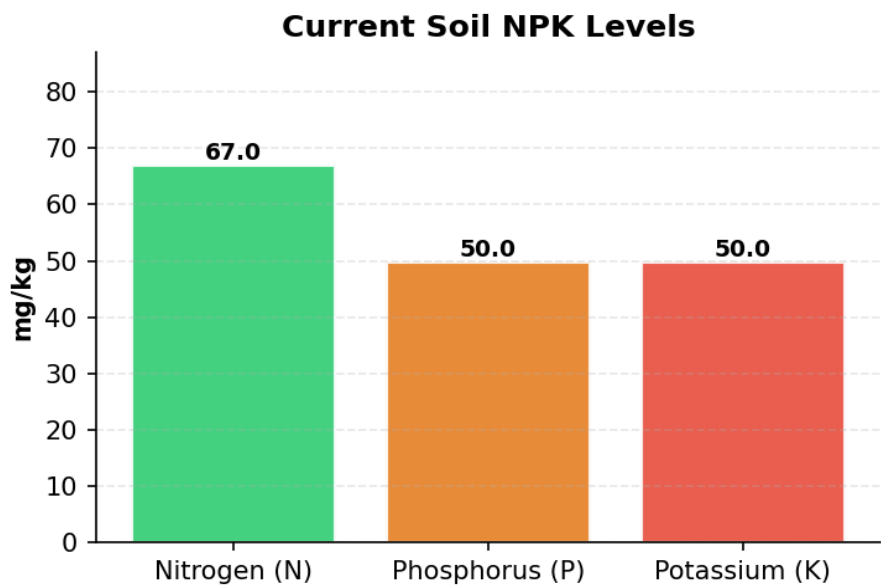
Alternative Crop Options

Rank	Crop	Confidence (%)
1	Banana	13.5%
2	Pigeonpeas	11.2%

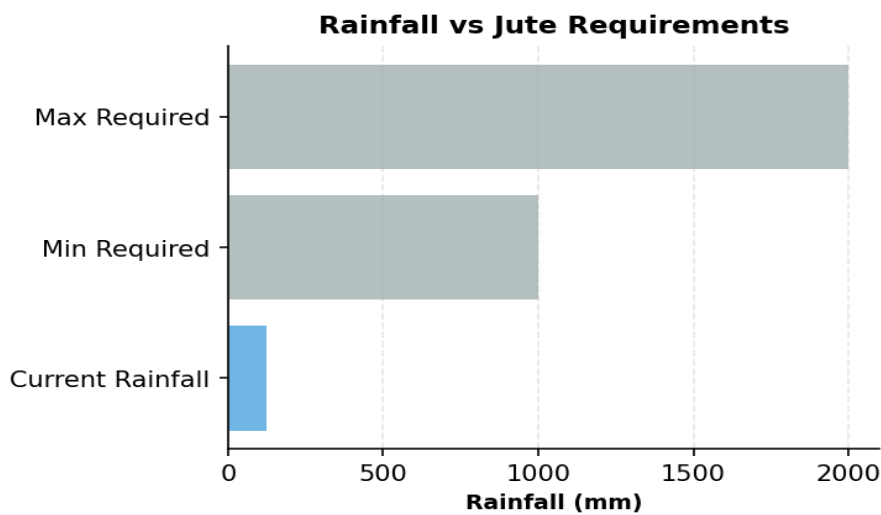
Soil NPK Levels Visualization

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Rainfall Analysis



Soil Treatment Recommendations

- Soil pH (6.5) is within optimal range (6.0-7.5) for jute.
- Nitrogen (67.0 mg/kg) is within optimal range for jute.
- Phosphorus (50.0 mg/kg) is excessive for jute. Optimal range: 20-40. High P can interfere with micronutrient uptake.
- Potassium (50.0 mg/kg) is excessive for jute. Optimal range: 20-40. High K can cause magnesium deficiency.
- Rich in nutrients, good water retention. Suitable for most crops.

Fertilizer Recommendations

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■ Soil NPK levels are already optimal for jute.

Maintain soil health with organic compost and crop rotation.

Organic Alternatives:

- Compost/FYM: 5-10 tons/acre (provides N, P, K)
- Green Manure: Leguminous crops (fixes N)
- Biofertilizers: Rhizobium (N), PSB (P), VAM (P)

Crop-Specific Tips for Jute

Requires high humidity and consistent moisture. Harvest before flowering for best fiber.

Additional Notes

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.